

Final Project Report Template

Project Title: Strategic Workforce Analytics and Attrition Insights

1. Introduction

1.1. Project overviews :

The **Strategic Workforce Analytics and Attrition Insights** project analyzes employee data to understand workforce characteristics and employee attrition patterns. Power BI was used to transform employee data into interactive reports and dashboards. The project focuses on attrition, work mode, department, job role, salary growth, burnout, performance, career level, training, and employee retention factors.

1.2. Objectives :

- Analyze overall employee attrition.
- Identify departments and job roles with higher attrition rates.
- Analyze the relationship between work mode and attrition.
- Study the impact of overtime, burnout, performance, and salary growth.
- Analyze employee retention across education, career level, tenure, and marital status.
- Provide interactive Power BI reports for HR decision-making.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement :

The organization has **5,000 employees**, of which **895 employees have left**, resulting in an overall attrition rate of approximately **18%**. The project aims to understand the major patterns associated with employee attrition, including compensation growth, workload, burnout, employee engagement, career progression, performance, and work arrangement.

2.2. Project Proposal (Proposed Solution) :

A Power BI-based **Strategic Workforce Analytics and Attrition Insights** solution was proposed. The solution uses employee data to create interactive reports containing KPIs, charts, filters, and analytical visuals. The reports help identify attrition patterns and provide HR with a clear view of workforce and retention-related factors.

2.3. Initial Project Planning :

The project was completed individually and divided into the following activities:

Phase	Activity
Phase 1	Project initialization and problem definition
Phase 2	Data collection and understanding
Phase 3	Data quality checking and preprocessing
Phase 4	Business question identification
Phase 5	Power BI visualization development
Phase 6	Dashboard and report creation
Phase 7	Testing and validation
Phase 8	Final observations and documentation

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified:

The project uses an employee attrition dataset containing employee-related information such as department, job role, work mode, job level, overtime, burnout, performance, salary growth, education level, career information, and attrition status.

The collected data was imported into Power BI for analysis and visualization.

3.2. Data Quality Report :

Data Quality Issue	Severity	Resolution Plan
Missing values	Medium	Checked and handled where required
Duplicate employee records	High	Checked employee records and removed/avoided duplicate records
Inconsistent categorical values	Medium	Standardized category values
Incorrect data types	Medium	Converted columns to appropriate data types
Unclear column names	Low	Reviewed and used meaningful field names for analysis
Outliers in numerical fields	Medium	Reviewed during data exploration

3.3. Data Exploration and Preprocessing :

The dataset was explored to understand the employee attributes and identify useful variables for analysis. Relevant fields were selected for calculating attrition rate, employee count, AI adoption, average salary-related values, years at company, burnout, engagement, manager support, performance, and other workforce indicators.

The data was then prepared for Power BI visualization by checking data types, categories, duplicates, and data quality issues.

4. Data Visualization

4.1. Framing Business Questions:

The following business questions were considered:

1. What is the overall employee attrition rate?
2. Which departments have higher attrition rates?
3. Which job roles show higher attrition?
4. How does work mode affect attrition?
5. How does salary growth relate to attrition?
6. How does overtime affect employee attrition?
7. How does employee performance relate to attrition?
8. How does burnout relate to attrition?
9. How does attrition vary across career levels?
10. How does employee retention vary across marital status?
11. What is the relationship between AI tool usage and employee attrition?

4.2. Developing Visualizations :

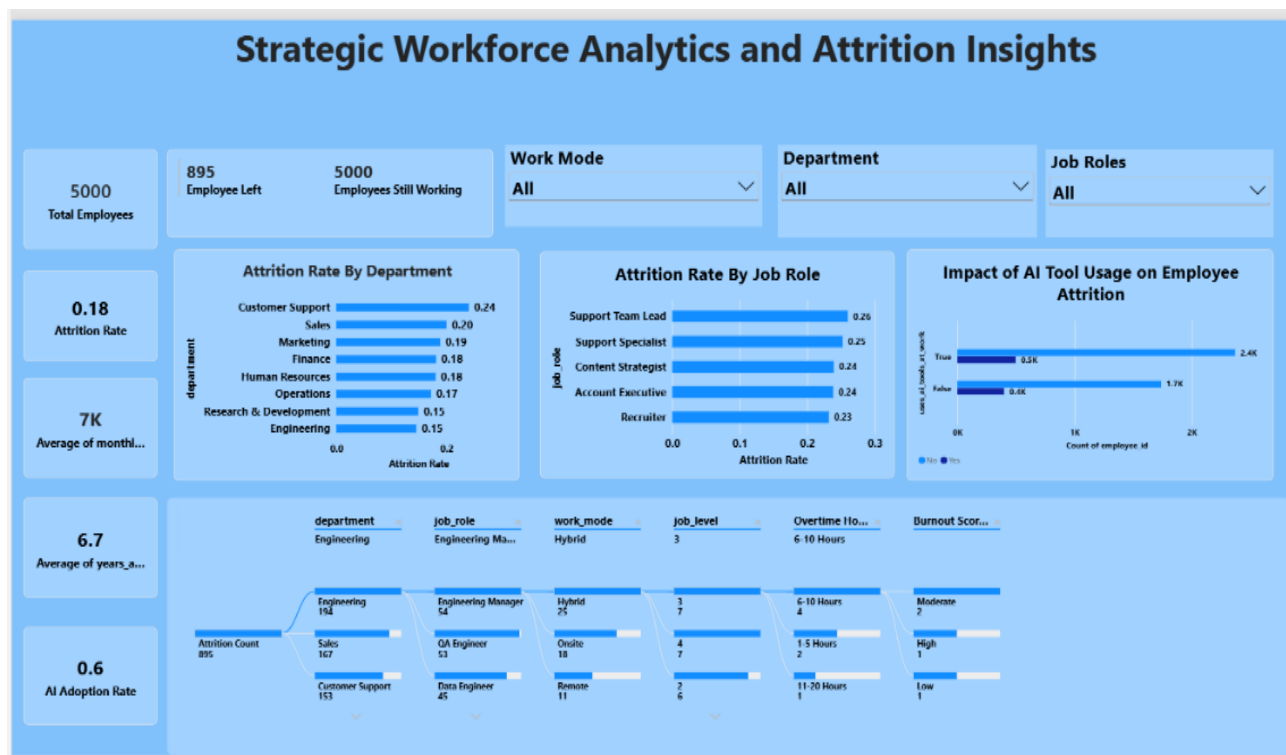
Power BI visualizations were developed using:

- KPI cards
- Bar charts
- Stacked bar charts
- Line charts
- Donut charts
- Scatter chart
- Slicers
- Decomposition/tree-style analysis

Interactive filters were provided for attributes such as **department**, **job role**, **work mode**, **education level**, **gender**, and **marital status**.

5. Dashboard

5.1. Dashboard Design Screenshot with Insights: The main dashboard, **Strategic Workforce Analytics and Attrition Insights**, provides an overview of workforce size and attrition.



Key Insights

- Total employees: **5,000**
- Employees left: **895**
- Employee attrition rate: approximately **18%**
- Customer Support shows the highest departmental attrition rate at approximately **24%**.
- Support Team Lead shows the highest attrition rate among the displayed job roles at approximately **26%**.
- AI tool usage is compared with employee attrition using employee counts.
- The dashboard provides filters for **work mode**, **department**, and **job role**.

- The dashboard also includes analysis of department, job role, work mode, job level, overtime, and burnout

6. Report

6.1. Report Design Screenshot with Insights: The project contains multiple Power BI reports that provide detailed analysis of employee retention, workforce performance, and career growth.

Report 1 – Employee Retention Performance Intelligence

This report focuses on employee retention and factors associated with attrition.

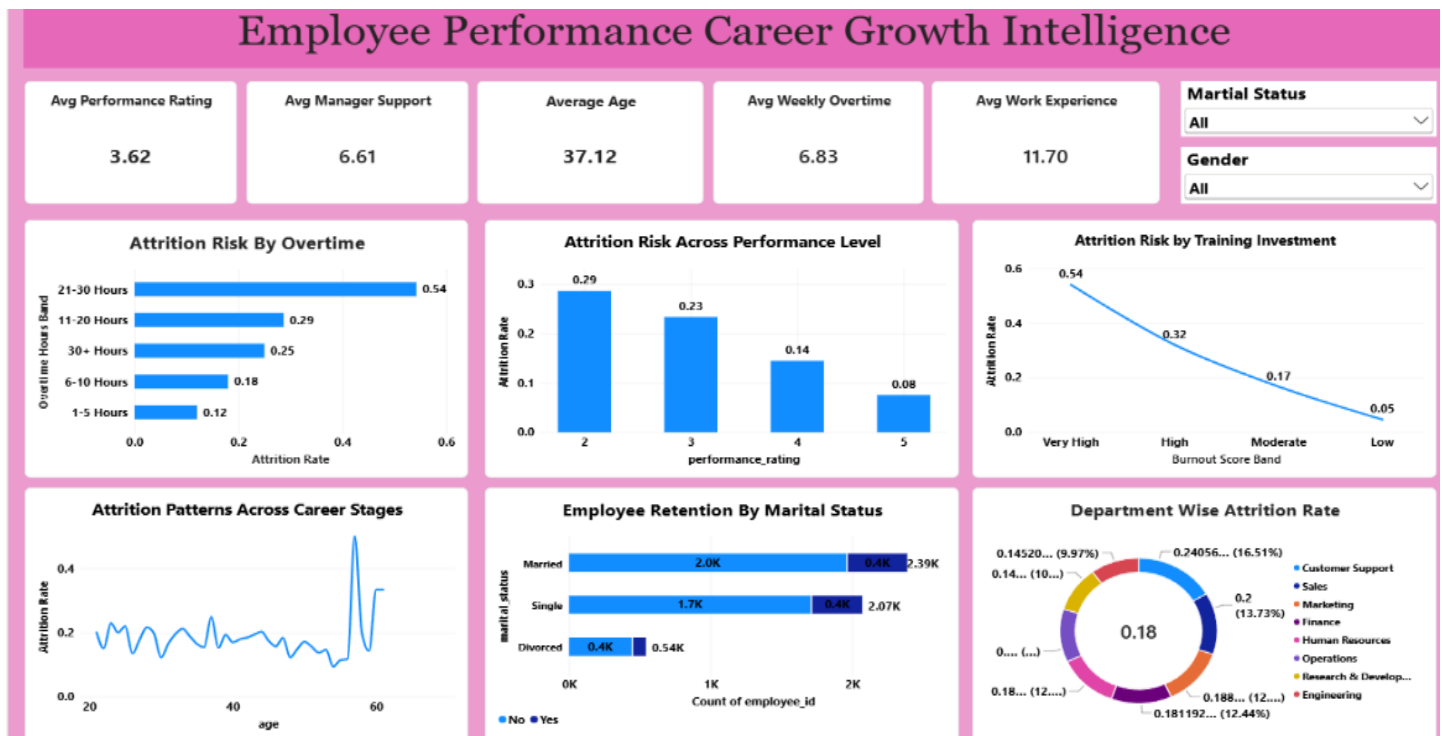


Key Insights

- Average Burnout Score: **4.03**
- Average Work-Life Balance Score: **6.13**
- Employee Engagement Index: **6.51**
- Manager Support Score: **6.61**
- Salary growth band **0–5%** has an attrition rate of approximately **30%**.
- Attrition varies across educational backgrounds.
- Burnout risk is compared across onsite, hybrid, and remote work modes.
- Attrition patterns are analyzed across employee tenure.
- Attrition is also compared across workplace models and career levels.

Report 2 – Employee Performance Career Growth Intelligence

- This report focuses on employee performance, overtime, training, career stages, and employee characteristics.



Key Insights

- Average Performance Rating: **3.62**
- Average Manager Support: **6.61**
- Average Age: **37.12**
- Average Weekly Overtime: **6.83 hours**
- Average Work Experience: **11.70 years**
- Employees working **21–30 overtime hours** show an attrition rate of approximately **54%**.
- Attrition rate decreases as performance rating increases from the displayed levels.
- Attrition varies across burnout score bands.
- Employee retention is analyzed across marital status.
- Department-wise attrition rates are displayed using a donut chart.
- Career-stage attrition patterns are shown using a line chart.

. Performance Testing

7.1 Utilization of Data filters :

Interactive slicers and filters were used to allow users to analyze employee data based on different categories.

Filters include:

- Department
- Job Role
- Work Mode
- Education Level

- Gender
- Marital Status

These filters allow users to dynamically view different employee segments.

7.2 No of Calculation Columns/Measures:

The Power BI model uses calculated measures and columns to derive important analytical values such as:

- Total Employees
- Employee Left
- Employees Still Working
- Attrition Rate
- AI Adoption Rate
- Average Monthly Income
- Average Years at Company
- Average Burnout Score
- Average Work-Life Balance Score
- Employee Engagement Index
- Manager Support Score
- Average Performance Rating
- Average Age
- Average Weekly Overtime
- Average Work Experience

7.3 No of Visualization :

The reports use multiple visualization types, including:

- KPI cards
- Bar charts
- Stacked bar charts
- Line charts
- Donut chart
- Scatter chart
- Interactive slicers
- Decomposition/tree analysis

These visualizations provide different views of employee attrition and retention patterns.

8. Conclusion/Observation:

The project provides an interactive Power BI solution for analyzing employee attrition and workforce characteristics. The analysis shows that attrition varies across departments, job roles, salary growth bands, overtime levels, burnout levels, workplace models, performance ratings, and career levels.

The dashboards provide HR stakeholders with a consolidated view of employee attrition and retention-related factors. Interactive filters make it possible to analyze specific employee groups and identify patterns within different workforce segments.

9. Future Scope :

The project can be further enhanced by:

- Adding predictive employee attrition models.

- Using machine learning to identify high-risk employees.
- Adding time-based attrition forecasting.
- Integrating real-time HR data.
- Adding automated HR alerts for high-risk segments.
- Including additional employee engagement and compensation data.
- Developing a mobile-friendly Power BI dashboard.
- Adding more advanced AI-based workforce recommendations.

10. Appendix

10.1. Source Code (if any): No separate source code was used. The project was developed using Microsoft Power BI, including Power Query for data preprocessing, calculated columns and measures using DAX, and Power BI visualizations for dashboard and report development.

10.2. GitHub & Project Demo Link:

GitHub link:

https://github.com/bhavanigudikanti/Strategic_Employee_Attrition_Intelligence_Dashbord